**St. Philomena's College (Autonomous), Mysore**

**PG Department of Biochemistry**

**Question Bank (Revised Curriculum 2020-21 onwards)**

**First Year- First Semester ( 2020-21 Batch)**

**Course Title (Paper Title): Clinical Biochemistry(SC) QP Code - 84132**

| Sl. No | Unit | Questions | Marks |
|---|---|---|---|
| 1 | 1 | Define cell death. | 2 |
| 2 | 1 | Define hyperplasia & dysplasia. | 2 |
| 3 | 1 | Define Health and Disease. | 2 |
| 4 | 1 | Enlist the biological agents responsible for cell death. | 2 |
| 5 | 1 | Enlist the chemical agents responsible for cell death. | 2 |
| 6 | 1 | Define cell injury. Name the agents responsible for it. | 2 |
| 7 | 1 | Define Atrophy & Hypertrophy | 2 |
| 8 | 1 | How metaplasia differs from dysplasia? | 2 |
| 9 | 1 | Enlist the pathological factors responsible for cell death. | 2 |
| 10 | 1 | What is cell death? | 2 |
| 11 | 1 | Mention the importance of isoenzymes in clinical diagnosis. | 2 |
| 12 | 1 | Name any two isoenzymic forms of LDH with clinical significance. | 2 |
| 13 | 1 | What are isozymes? Give an example | 2 |
| 14 | 1 | How apoptosis differs from that of Necrosis? | 2 |
| 15 | 1 | Name any two isoenzymic forms of CK with clinical significance. | 2 |
| 16 | 1 | What are Transaminases? Give an example. | 2 |
| 17 | 1 | Give the clinical importance of Serum Amylase. | 2 |
| 18 | 1 | Give the clinical importance of Acid Phosphatases. | 2 |
| 19 | 1 | Give the clinical importance of Alkaline Phosphatases. | 2 |
| 20 | 1 | What are isoenzymes? Give its importance in clinical diagnosis. | 2 |
| 21 | 1 | Mention the importance of isoenzymes in clinical diagnosis. | 2 |
| 22 | 2 | Define Hypoglycemia & Hypoglycemia. | 2 |

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| 23 | 2 | Mention the types of diabetes | 2 |
| 24 | 2 | Define homeostasis | 2 |
| 25 | 2 | What are advanced glycation products? Give its significance | 2 |
| 26 | 2 | Mention the clinical symptoms seen in fructose intolerance. | 2 |
| 27 | 2 | Define blood sugar homeostasis. | 2 |
| 28 | 2 | Mention the role of insulin in blood sugar homeostasis. | 2 |
| 29 | 2 | Mention the role of glucagon in blood sugar homeostasis. | 2 |
| 30 | 2 | What is HbA1c? Give its importance. | 2 |
| 31 | 2 | Give the classification of Diabetes Mellitus. | 2 |
| 32 | 2 | Give the significance of GTT. | 2 |
| 33 | 2 | What is lactose intolerance? How to overcome it. | 2 |
| 34 | 2 | What is reverse cholesterol transport? | 2 |
| 35 | 2 | What is Galactosemia? How to overcome it. | 2 |
| 36 | 2 | What is glycated hemoglobin? How it is used in diagnosis of DM. | 2 |
| 37 | 2 | What is petosuria? How to overcome it. | 2 |
| 38 | 2 | List the genetic defects seen in patients suffering from lactose intolerance | 2 |
| 39 | 2 | What is Alcoholic fatty liver? | 2 |
| 40 | 2 | What is fatty liver? How is it caused? | 2 |
| 41 | 2 | What is the molecular defect seen in Niemann-pick disease? | 2 |
| 42 | 2 | What are foam cells? How are they made? | 2 |
| 43 | 2 | Define Atherosclerosis | 2 |
| 44 | 2 | Mention the clinical symptoms seen in Tay-Sach's Disease. | 2 |
| 45 | 2 | Mention the etiology and symptoms of Gaucher’s Disease. | 2 |
| 46 | 2 | Differentiate between LDL and Ox-LDL. | 2 |
| 47 | 3 | Enlist the clinical manifestations seen in alkaptonuria. | 2 |
| 48 | 3 | Distinguish between Hypouricemia and Hyperuricemia. | 2 |
| 49 | 3 | Name the molecular defect and symptoms seen in alkaptonuria. | 2 |

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| 50 | 3 | What causes Orotic aciduria? How to overcome it. | 2 |
| 51 | 3 | What causes Gout? How do you diagnose it. | 2 |
| 52 | 3 | What is cystinuria? | 2 |
| 53 | 3 | Enlist the clinical manifestations seen in albinism. | 2 |
| 54 | 3 | What causes maple syrup urine disease? | 2 |
| 55 | 3 | What is the molecular defect seen in alkaptonuria? | 2 |
| 56 | 3 | What is the molecular defect seen in Phenylketonuria? | 2 |
| 57 | 3 | List the clinical manifestations seen in Phenylketonuria. | 2 |
| 58 | 3 | What is CRP test? Give its importance. | 2 |
| 59 | 3 | What is Rheumatoid Arthritis test? Give its importance. | 2 |
| 60 | 4 | How gall stone are formed? How they differ from renal stones. | 2 |
| 61 | 4 | What is Jaundice? What causes it? | 2 |
| 62 | 4 | What is liver Cirrhosis? | 2 |
| 63 | 4 | What are mineralcorticoids? Give an example. | 2 |
| 64 | 4 | What are gall stones? How are they formed? | 2 |
| 65 | 4 | What is steatorrhea? How it differs from Diarrhea. | 2 |
| 66 | 4 | Enlist the causes for gastric ulcer. | 2 |
| 67 | 4 | What is stimulation test? Give its significance. | 2 |
| 68 | 4 | Name the different types of peptic ulcers. | 2 |
| 69 | 4 | What is Diarrhea? How it is different from Stearrhea? | 2 |
| 70 | 4 | Mention the causes and symptoms of Steatorrhea. | 2 |
| 71 | 5 | Define Glomerular Filtration Rate. | 2 |
| 72 | 5 | What are Creatinine clearance tests? | 2 |
| 73 | 5 | Enlist the abnormal constituents of urine. | 2 |
| 74 | 6 | Mention the clinical symptoms seen in Grave's Disease | 2 |
| 75 | 6 | Mention the clinical symptoms seen in Hashimoto's Disease. | 2 |
| 76 | 6 | Mention the functioning of Hypothalamus-Pituitary-Adrenal axis. | 2 |

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| 77 | 6 | Give the symptoms and therapy for hypothyroidism. | 2 |
| 78 | 6 | Mention the symptoms and therapy for hyperthyroidism. | 2 |
| 79 | 6 | Distinguish between Acromegaly and Gigantism. | 2 |
| 80 | 6 | Mention the clinical symptoms seen in Addison's Disease | 2 |
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| 1 | 1 | Elaborate on cell injury and cellular adaptations. | 5 |
| 2 | 1 | Write short note on Necrosis. | 5 |
| 3 | 1 | How are clinical enzymes useful in diagnosis of diseases? Explain with examples. | 5 |
| 4 | 1 | Explain the physiological causes responsible for cell death. | 5 |
| 5 | 1 | Explain cell injury and cellular adaptations. | 5 |
| 6 | 1 | Give an account on normal & pathological changes affecting cells in the body. | 5 |
| 7 | 1 | Write short note on Necrosis. | 5 |
| 8 | 1 | Explain the application of Transaminases in detection of a disease | 5 |
| 9 | 1 | Explain the mechanism of elevated enzyme activities in disease diagnosis. | 5 |
| 10 | 1 | How LDH is used in prognosis & diagnosis of a disease? Explain | 5 |
| 11 | 1 | Explain the role of serum enzymes in clinical diagnosis. | 5 |
| 12 | 2 | Comment on Galactosemia & petosuria. | 5 |
| 13 | 2 | Comment on fructosuria & fructose intolerance | 5 |
| 14 | 2 | Discuss on glycogen storage diseases. | 5 |
| 15 | 2 | Give an account on glycogen storage diseases. | 5 |
| 16 | 2 | Discuss on classification, metabolic abnormalities, diagnosis and management. for Diabetes Mellitus | 5 |
| 17 | 2 | Discuss on complications &laboratory investigations for Diabetes Mellitus | 5 |
| 18 | 2 | Give an account on glycogen storage diseases. | 5 |
| 19 | 2 | Discuss on types, metabolic abnormalities s &laboratory investigations for Diabetes Mellitus | 5 |
| 20 | 2 | Comment on Galactosemia & petosuria. | 5 |

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| 21 | 2 | Give the classification, etiology, clinical manifestation and treatment of Diabetes Mellitus. | 5 |
| 22 | 2 | Explain congenital errors of carbohydrate metabolism with an suitable example. | 5 |
| 23 | 2 | Give the symptoms, molecular defects and treatment options for the following metabolic disease: Lactate intolerance and Petosuria. | 5 |
| 24 | 2 | Briefly explain the general physiological markers ad phenotypes for and classify Diabetes. Mention the treatment plan for each class. | 5 |
| 25 | 2 | Give the classification an major functions of lipoprotein. Add a note on reverse cholesterol transport. | 5 |
| 26 | 2 | Enlist the risk factors contributing to the pathogenesis of cardiovascular diseases. | 5 |
| 27 | 2 | Describe Niemann-pick disease. | 5 |
| 28 | 2 | Explain the prognosis and risk factors of Atherosclerosis. | 5 |
| 29 | 2 | Give an account on Lipoproteinemias. | 5 |
| 30 | 2 | Explain different types of Lipoproteinemias. Add a note on Atherosclerosis. | 5 |
| 31 | 2 | Explain Tay-Sach's Disease. | 5 |
| 32 | 2 | Explain Gaucher’s Disease. | 5 |
| 33 | 2 | Write a brief note on causes and treatment for cardiovascular diseases. | 5 |
| 34 | 3 | Give the symptoms, molecular defects and treatment options for the following metabolic disease: Phenylketonuria | 5 |
| 35 | 3 | Give the symptoms, molecular defects and treatment options for the following metabolic disease: Gout | 5 |
| 36 | 3 | Give the symptoms, molecular defects and treatment options for the following metabolic disease: alkaptonuria | 5 |
| 37 | 3 | Give the symptoms, molecular defects and treatment options for the following metabolic disease: albinism | 5 |
| 38 | 3 | Give the symptoms, molecular defects and treatment options for the following metabolic disease: maple syrup urine disease | 5 |
| 39 | 3 | Explain Rheumatoid arthritis (RA) test & CRP test | 5 |
| 40 | 4 | Write short note on types, treatment and diagnosis for Jaundice | 5 |
| 41 | 4 | Which are the tests to be done to assess the proper functioning of Liver? | 5 |

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| | | Describe in detail. | |
| 42 | 4 | What are liver function tests? Explain in detail. | 5 |
| 43 | 4 | Explain the formation of Bilirubin. | 5 |
| 44 | 4 | What are bile pigments? Explain the formation of Bilirubin. | 5 |
| 45 | 4 | Write short note on Jaundice. | 5 |
| 46 | 4 | How Bilirubin is an excellent diagnostic tool for monitoring liver function? | 5 |
| 47 | 4 | What are gastric function tests? Explain in detail. | 5 |
| 48 | 4 | List the differences and similarities between hypo and hyper secretion acidity. How do they affect Malabsorption syndrome. | 5 |
| 49 | 4 | Write short note on fractional gastric analysis and give its significance. | 5 |
| 50 | 4 | Write short note on Malabsorption syndrome. | 5 |
| 51 | 4 | Explain Zollinger Ellison syndrome. | 5 |
| 52 | 4 | Explain the tests related to excretory & synthetic function of liver. | 5 |
| 53 | 4 | Explain the tests related to detoxifying and metabolic function of liver. | 5 |
| 54 | 4 | Write short note on Hepatitis. | 5 |
| 55 | 4 | What are gall stones? Explain their formation. | 5 |
| 56 | 5 | How renal functions are assessed in clinical laboratory? | 5 |
| 57 | 5 | Which are the tests to be done to assess the proper functioning of Kidney? Describe in detail. | 5 |
| 58 | 5 | Give an account on Abnormal constituents of urine. | 5 |
| 59 | 5 | What are kidney function tests? Explain in detail. | 5 |
| 60 | 5 | What are Glomerular function tests? Explain in detail. | 5 |
| 61 | 5 | What are tubular function tests? Explain in detail. | 5 |
| 62 | 5 | What are kidney clearance tests? How are they used as diagnostic tools? | 5 |
| 63 | 5 | How renal function can be assessed in clinical laboratory? | 5 |
| 64 | 6 | Choose an endocrine disorder of your choice, explain the molecular defect associated with it and how it diagnosed in clinical laboratory. | 5 |
| 65 | 6 | Write an explanatory note on hypo and hyper secretion of hormones. | 5 |

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| 66 | 6 | Explain the causes, defects and diagnosis of any one disorder of your choice related to pituitary gland. | 5 |
| 67 | 6 | Write short note on Grave's Disease. | 5 |
| 68 | 6 | Choose an endocrine disorder related to pituitary gland; Explain the molecular defect associated with it and how it diagnosed in clinical laboratory. | 5 |
| 69 | 6 | Explain the causes, defects and diagnosis of any one disorder of your choice related to thyroid gland. | 5 |
| 70 | 6 | Explain the causes, defects and diagnosis of any one disorder of your choice related to Adrenal Gland. | 5 |
| 71 | 6 | Give an account on hypo and hyper secretion of hormones with suitable example. | 5 |
| 72 | 6 | Explain how hypo and hyper secretion of growth hormone affect the system? How are they diagnosed? | 5 |
| 73 | 6 | Explain Thyroid disorders | 5 |
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| 1 | 1 | How Clinical enzymes are useful in diagnosis of diseases? Explain in detail. | 10 |
| 2 | 1 | Explain the mechanism of elevated enzyme activities in prognosis and diagnosis of disease | 10 |
| 3 | 1 | Give an account on application of serum enzymes in prognosis and diagnosis of a disease. | 10 |
| 4 | 2 | How blood sugar level is regulated? Explain. | 10 |
| 5 | 2 | Give an account on role of hormones and tissues in maintaining blood glucose homeostasis. | 10 |
| 6 | 2 | Give an account on metabolic abnormalities, diagnosis, management & Laboratory Investigations for Diabetes Mellitus | 10 |
| 7 | 2 | Give an account on congenital errors of carbohydrate metabolism. | 10 |
| 8 | 2 | Explain the Lipid storage diseases. | 10 |
| 9 | 2 | Elaborate on Risk Factors , Pathogenesis , Diagnosis And Prognosis of Atherosclerosis . | 10 |
| 10 | 3 | Give the symptoms, molecular defects and treatment options for the following metabolic disease: Gout and maple syrup urine disease | 10 |
| 11 | 3 | Explain congenital errors of Amino acid metabolism with an suitable example. | 10 |

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| 12 | 3 | Give an account on congenital errors of Amino acid metabolism. | 10 |
| 13 | 3 | Give the symptoms, molecular defects and treatment options for the following metabolic disease: Gout and Phenylketonuria. | 10 |
| 14 | 3 | Explain congenital errors of Nucleic acid metabolism with a suitable example. | 10 |
| 15 | 3 | Give an account on congenital errors of Nucleic acid metabolism. | 10 |
| 16 | 3 | Give the symptoms, molecular defects and treatment options for the following metabolic disease: Phenylketonuria and Hyperuricemia | 10 |
| 17 | 4 | Explain the liver function tests in detail. | 10 |
| 18 | 4 | Give an account on Malabsorption Syndrome & Zollinger Ellison syndrome | 10 |
| 19 | 5 | Explain the biochemical findings in detection of kidney disorders | 10 |
| 20 | 5 | Give an account on Glomerular & tubular function tests. | 10 |
| 21 | 6 | Choose an endocrine disorder related to thyroid gland; Explain the molecular defect associated with it and how it diagnosed in clinical laboratory | 10 |
| 22 | 6 | Choose an endocrine disorder related to pituitary gland; Explain the molecular defect associated with it and how it diagnosed in clinical laboratory | 10 |
| 23 | 6 | Choose an endocrine disorder related to adrenal gland; Explain the molecular defect associated with it and how it diagnosed in clinical laboratory. | 10 |
| 24 | 6 | Explain Graves disease & Addission's Disease | 10 |
| 25 | 6 | Explain thyroid diseases. | 10 |
| 26 | 6 | Explain the disorders related to pituitary gland. | 10 |
| 27 | 6 | Explain the concept of hypo & hyper secretion of hormones with suitable example. | 10 |
| 28 | 6 | Explain Hashimoto disease & Addission's Disease | 10 |
| 29 | 6 | Explain the concept of hypo & hyper secretion of hormones with pituitary hormones as an example. | 10 |
| 30 | 6 | Explain the concept of hypo & hyper secretion of hormones with thyroid hormones as an example. | 10 |

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| **Model Question Paper** | | | |
| **St. Philomena's College (Autonomous), Mysore** | | | |
| **I Semester M.Sc-Final Examination** | | | |
| **Subject: Biochemistry** | | | |
| **Title: Clinical Biochemistry (SC)** | | | |
| **Time: 3 Hours** | | | **Max Marks: 70** |
| **PART-A** | | | |
| **Answer any TEN of the following:** | | | **10x2=20** |
| **1.** | | Define hyperplasia & dysplasia. | **2** |
| **2.** | | Name any two isoenzymic forms of CK with clinical significance. | **2** |
| **3.** | | What are advanced glycation products? Give its significance | **2** |
| **4.** | | Define homeostasis. | **2** |
| **5.** | | What causes Orotic aciduria? | **2** |
| **6.** | | What is Hyperuricemia? | **2** |
| **7.** | | Enlist the causes for gastric ulcer. | **2** |
| **8.** | | What is stimulation test? Give its significance. | **2** |
| **9.** | | What are Creatinine clearance tests? | **2** |
| **10.** | | Define Glomerular Filtration Rate. | **2** |
| **11.** | | Distinguish between Acromegaly and Gigantism. | **2** |
| **12.** | | Mention the clinical symptoms seen in Addison's Disease | **2** |
| **PART-B** | | | |
| **Answer any SIX questions:** | | | **6x5=30** |
| **13.** | | Write short note on Necrosis. | **5** |
| **14.** | | Discuss on glycogen storage diseases. | **5** |
| **15.** | | Give the symptoms, molecular defects and treatment options for the following metabolic disease: Gout & Galactosemia. | **5** |
| **16.** | | Write short note on Jaundice. | **5** |

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| **17.** | | Give an account on Abnormal constituents of urine. | **5** |
| **18.** | | Write an explanatory note on hypo and hyper secretion of hormones. | **5** |
| **19.** | | Explain Rheumatoid arthritis (RA) test & CRP test | **5** |
| **20.** | | Explain different types of Lipoproteinemias. Add a note on Atherosclerosis. | **5** |
| | | **PART-C** | |
| **Answer any TWO questions:** | | | **2x10=20** |
| **21.** | | Give an account on role of hormones and tissues in maintaining blood glucose homeostasis. | **10** |
| **22.** | | Give an account on congenital errors of amino acid metabolism. | **10** |
| **23.** | | Give an account on Malabsorption Syndrome & Zollinger Ellison syndrome | **10** |